TaskDesk Developer Guide

Oct 11, 2026 · @Pradeep

Overview

TaskDesk is the internal work app for the Search Engineers team: project settings create recurring tasks automatically, people work and discuss them, and the admin sees attendance, leave, clients and monthly reports in one place. This guide explains the logic behind every module so the team can build the production app from the approved clickable preview.

Source of truth. The clickable preview (Version 10) defines the behaviour. Where this guide and the preview differ, the preview wins and this guide should be corrected.

Scope of version 1

- Sign in, roles (Admin, Lead, Member) and people management
- Projects with recurring task rules and service templates
- Tasks: one time and recurring, subtasks, dependencies, approvals, comments with mentions, time logs
- Calendar: weekly offs, holidays, restricted holidays, leave requests approved only by the admin
- Attendance from sign in and sign out, activity logs per profile
- Notifications, admin announcements, a personal dashboard
- Office pages: Policies with acknowledgement, People directory, Tools list, Credentials
- Clients (admin only): pipeline stages, multi service pricing, payments, domain, hosting and AMC renewals
- Reports by day, week, month and quarter with CSV export

Glossary

Term	Meaning
Rule	A recurring task definition on a project (what, who, how often)
Occurrence	One calendar date a rule fires on, before holiday shifting

Term	Meaning
Due date	The date the generated task is due, after holiday shifting
Lookahead	How many days ahead tasks are created, 7 by default
Blocker	An open linked task or a pending approval that stops a task being marked done
Session	One sign in to sign out span, used for attendance
Activity	One logged action by a person, shown on task history and profile logs
Live stage	Client stages Active and Onboarding, which count toward revenue
Access level	Hidden, Login only or Full access, set per credential for leads and members

Tech stack and architecture

TaskDesk is one Next.js 15 app (App Router, React 19, server actions) on a single server, with Prisma 6 over SQLite for a small team or PostgreSQL when it grows. There is no separate API service: pages read through server components and write through server actions, and one cron endpoint runs the daily jobs.

Layer	Choice	Notes
Framework	Next.js 15.5, React 19, TypeScript 5	App Router, server actions for every write
Database	Prisma 6.19 with the client engine and driver adapters	libSQL adapter for SQLite, pg adapter for PostgreSQL
Auth	bcryptjs password hashes, jose signed session cookie	Middleware guards every page except login
Styling	Tailwind CSS 3.4	Brand tokens: red #EF4045, dark red #C9242A, black #141416, white background

Layer	Choice	Notes
Fonts	Outfit for headings, Figtree for body	Loaded from Google Fonts
Jobs	One cron call to api/cron/generate each morning	Protected by CRON_SECRET
Time zone	Asia/Kolkata for every date decision	Set by APP_TIMEZONE

Folder layout (existing source, to be extended)

Path	Holds
prisma/schema.prisma	The data model (section Data model lists every entity)
prisma/seed.ts	First admin account, optional demo data when SEED_DEMO is true
src/lib/db.ts	Prisma client with the right adapter for DATABASE_URL
src/lib/auth.ts, session.ts	Password check, cookie signing, current user lookup
src/lib/permissions.ts	Every can… rule in one file (see Roles and permissions)
src/lib/recurrence.ts	Occurrence matching, holiday shifting, generation
src/lib/dates.ts	Day keys, working day checks, time zone helpers
src/lib/activity.ts	The act() helper that writes the activity log
src/lib/report.ts, queries.ts	Report numbers and shared read queries
src/lib/templates.ts	Service templates (SEO, PPC, Content, Web development, Reporting)
src/app/(app)/…	Signed in pages, one folder per section, actions.ts beside each page
src/app/api/cron/generate	Daily job endpoint
src/app/api/reports/csv	CSV export
scripts folder, generate tasks script	Same daily job, run from the command line

New folders to add, matching the preview: notifications, announcements, calendar, leave, attendance, people (with people/[id] for profiles), policies, tools, credentials, clients and clients/[id], plus src/lib/notify.ts, src/lib/calendar.ts, src/lib/crypto.ts and src/lib/clients.ts.

Rules every developer follows

1. Every write is a server action that loads the current user, checks permissions in src/lib/permissions.ts, writes inside one Prisma transaction, then calls act() and notify() where this guide says so.

2. Never trust the browser for role or ownership. Hidden buttons are only a convenience.
3. Dates without time are stored as day keys (text, year month day) in the team time zone. Timestamps are stored in UTC and shown in Asia/Kolkata.
4. Business logic lives in src/lib, never inside a page, so the cron job and the pages share it.

Roles and permissions

There are three roles: Admin sees and changes everything, Lead manages the projects they lead and the members who report to them, Member works their own tasks. Put every rule below in src/lib/permissions.ts and call it from both the page (to hide buttons) and the server action (to refuse the write).

Action	Admin	Lead	Member
See all tasks, projects, team, reports	Yes	Yes	Own tasks and projects they are on
Create a project and its recurring rules	Yes	Yes, becomes its lead	No
Edit or pause rules on a project	Yes	Only projects they lead	No
Create a one time task	Yes	Yes	Yes, for themselves
Edit a task	Yes	If they lead its project	If assignee, creator or collaborator
Reassign a task	Yes	If they lead its project, or task has no project	If they created it
Delete a task	Yes	No, use Cancelled	No, use Cancelled
Approve or reject leave	Yes	No	No
Add people	Leads and members	Members only, who then report to them	No

Action	Admin	Lead	Member
Edit a person	Anyone, including role and active	Members who report to them or whom they added, details only	Own phone and location
Reset a password	Anyone	Members they can edit	No
Share announcements	Yes	No	No
Edit policies, holidays, weekly off, leave allowance	Yes	No	No
Clients, pricing, payments, renewals	Yes	No	No
Credentials	Add, edit, set access	See what the admin shares	See what the admin shares
Activity log page	Everyone, including client entries	Own and every member's, client entries hidden	Own only

Key functions (names from the preview)

- canManage(user, project): admin, or a lead whose id is project.leadId
- canEdit(user, task): admin, assignee, creator, collaborator, or canManage of its project
- canReassign(user, task): admin, canManage of its project, a lead on a task with no project, or the creator
- canDelete(user): admin only
- canApprove(user, leave): admin only, never their own leave (admin leave is recorded as approved directly)
- canAddPeople(user) and addableRoles(user): admin adds Lead or Member, lead adds Member
- canEditPerson(user, person): admin, or a lead when the person is a Member who reports to them or was added by them
- credLevel(user, credential): admin always FULL, otherwise credential.access for the user's role

Guards to keep

- An admin cannot remove their own admin role or deactivate themselves.
- A lead cannot create a lead or admin even by editing the form; the server checks addableRoles.
- Inactive people cannot sign in, are hidden from assignee lists, and keep their history.

Data model

The existing schema has 10 models (User, Project, ProjectMember, RecurringRule, Task, TaskCollaborator, Comment, TimeLog, Activity, Setting); version 1 adds 20 more for the modules approved in the preview. Field names below match the preview so its logic ports directly. Every table also gets id (cuid), createdAt and updatedAt.

Work

Model	Key fields	Notes
User	name, email (unique), passwordHash, role (ADMIN, LEAD, MEMBER), title, phone, location, empType (ON_ROLE, ASSOCIATE), active, reportsToId, addedById, mustChangePassword	reportsToId and addedById point to User
Project	name, clientId, industry, line, status (ACTIVE, PAUSED, ARCHIVED), leadId, website, startDate, pocName, pocRole, pocPhone, pocEmail, comms (list)	POC and comms copy from the client when linked
ProjectMember	projectId, userId	Unique pair
RecurringRule	projectId, title, description, category, frequency (DAILY, WEEKLY, MONTHLY, QUARTERLY, ONCE), weekday, monthDay, skipWeekends, startDate, endDate, dueOffsetDays, assigneeId, priority, estimate, checklist (list), active, generatedThrough	generatedThrough is the preview's gen field

Model	Key fields	Notes
Task	title, description, projectId, assigneeId, createdById, ruleId, occurrenceDate, renewKey, frequency, category, status, priority, dueDate, estimate, spentMinutes, completedAt, cancelledAt	Unique (ruleId, occurrenceDate) and unique renewKey stop duplicates
Subtask	taskId, title, assigneeId, done, sortOrder	Rule checklist items become subtasks
TaskDependency	taskId, dependsOnId	Unique pair, no self link, no two way link
TaskApproval	taskId, approverId, requestedById, note, status (PENDING, APPROVED, REJECTED), decidedAt	Counts as a blocker while PENDING
TaskCollaborator	taskId, userId	Unique pair
Comment	taskId, userId, body	Mentions parsed on save
TimeLog	taskId, userId, minutes, workDate, note	Adds to task.spentMinutes
DayPlan	userId, date, items (taskId, slot MORNING, AFTERNOON or EVENING)	Plan my day widget, one per person per date

People and office

Model	Key fields	Notes
Session	userId, signedInAt, signedOutAt	Attendance source
Holiday	date, name, type (PUBLIC, RESTRICTED, COMPANY)	Restricted ones are optional personal days

Model	Key fields	Notes
Leave	userId, type (CASUAL, SICK, EARNED, COMPOFF, UNPAID, RH), fromDate, toDate, half, reason, status (PENDING, APPROVED, REJECTED, CANCELLED), decidedById, decidedAt	RH is a restricted holiday pick
Policy	title, body (markdown), updatedById, updatedOn	HR Policy, Leave Policy and more
PolicyAck	policyId, userId, ackedOn	Cleared when the policy body changes
Tool	name, category, url, username, passwordEnc, twofa, plan (FREE, PAID), access, visibility, notes	Category list includes Others
Credential	scope (CLIENT, INTERNAL), clientId, name, url, username, passwordEnc, notes, leadAccess, memberAccess, changedOn	Access is NONE, LOGIN or FULL
CredentialView	credentialId, userId, action, at	Every reveal and copy is logged
Notification	userId, kind, title, body, link, fromId, readAt	Kinds listed in Notifications
Announcement	title, body, byId	Read state in AnnouncementRead
AnnouncementRead	announcementId, userId, readAt	Hides the banner for that person
Activity	userId, action, detail, taskId, projectId, at	Already exists; also feeds profile logs
Setting	key, value (JSON)	sundayOff, satRule, leave allowances, rhLimit

Clients (admin only)

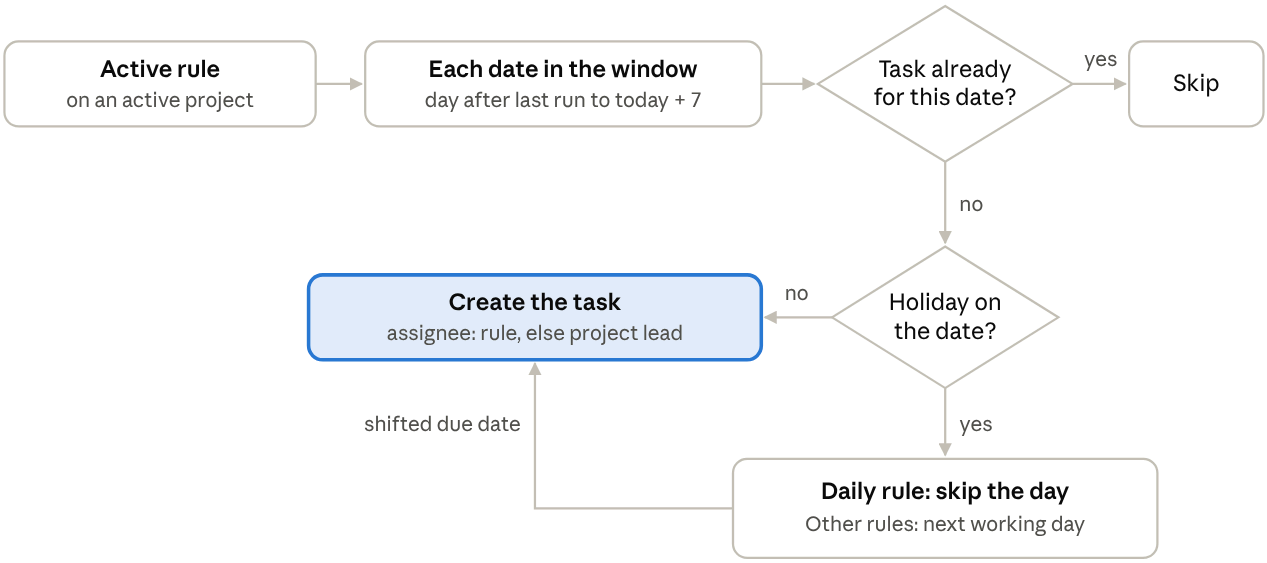
Model	Key fields	Notes
Client	name, industry, stage, ownerId, website, address, gst, pocName, pocRole, pocPhone, pocEmail, comms (list), onboardedOn, contractRenewal, notes	Stage list in Clients section
ClientContact	clientId, name, role, phone, email	Other contacts besides the POC
ClientService	clientId, line, type (RECURRING, ONE_TIME), price, cycle (MONTHLY, QUARTERLY, YEARLY), startDate, completedDate, item, provider, renewDate	One row per service, so one client can have many prices
Payment	clientId, amount, receivedOn, method, note	Feeds weekly revenue received
ClientNote	clientId, byId, on, text	Stage changes are logged here automatically

Store money as whole rupees in an integer column. Store passwords for tools and credentials encrypted (see Credentials and security), never as plain text.

Recurring task engine

Each rule on an active project creates real Task rows up to 7 days ahead, one per occurrence, and never twice for the same date. The same code runs from the daily cron job, when a rule is created or edited, and when holidays or weekly offs change.

Each rule date becomes exactly one task, shifted past holidays



After each run the rule remembers today + 7, so the next run starts the day after.

task generation · 2 checks per date

The duplicate check runs before the holiday shift, so a moved task is still matched to its original date on the next run.

Step 1. Does a rule fire on a date (matches)

Frequency	Fires when	Settings
DAILY	Every day, or every working day when skipWeekends is on	skipWeekends
WEEKLY	The weekday equals rule.weekday (0 Sunday to 6 Saturday, default Monday)	weekday
MONTHLY	The day of month equals monthDay, or the last day when the month is shorter	monthDay
QUARTERLY	Every third month counted from startDate, on monthDay with the same short month rule	monthDay, startDate
ONCE	Exactly one date: startDate plus dueOffsetDays	startDate, dueOffsetDays

A rule never fires before startDate or after endDate.

Step 2. Shift for holidays (scheduleDate)

- DAILY: if the date is a public or company holiday, skip it entirely. Daily checks do not pile up after a holiday.
- Every other frequency: if the date is a holiday, move forward one day at a time until the date is neither a holiday nor a weekly off. The task keeps occurrenceDate (the original date) and gets the shifted dueDate, and the task page shows "Moved from <date> for a holiday".
- Restricted holidays never move tasks; they are personal leave.
- A weekly off on the original date does not move WEEKLY, MONTHLY or QUARTERLY tasks. Only a holiday starts the move; once moving, weekly offs are skipped too.

Step 3. Generate (generate)

1. today = the current date in Asia/Kolkata; horizon = today plus LOOKAHEAD (7).
2. For each active rule whose project status is ACTIVE:
   a. ONCE: if already generated, skip; otherwise the window is its single date.
   b. Others: start from the day after generatedThrough, but never earlier than today minus 7 (so a server that was off for a while catches up one week, not months). A brand new rule starts at the later of startDate and today.
3. List occurrences in the window, skip any occurrenceDate that already has a task for this rule, shift each with scheduleDate, drop skipped ones.
4. Create the task: title, description, category, priority and estimate from the rule; assignee = rule.assigneeId, else the project lead; status TODO; checklist items become subtasks.
5. Set generatedThrough = horizon (ONCE: its date). Send an ASSIGNED notification to each assignee once per run, grouped ("5 new tasks for this week"), not one per task.

Step 4. Keep the future clean after edits

- A task is untouched when it is TODO, has no time logged, no comments, no ticked subtasks and no dependencies.
- Editing a rule (resetRuleFuture): delete only that rule's untouched tasks due after today, set generatedThrough to today, and generate again. Touched tasks stay as they are.
- Changing holidays or weekly offs (rescheduleAll): delete every untouched rule task due after today, pull generatedThrough back to today, and generate again for all rules.
- Pausing a rule or a project stops generation; existing tasks stay. Archiving a project also stops it.

Templates

New projects can add rules from templates per service line (SEO, Google Ads and PPC, Content and social, Web development, Client reporting). Each template holds title, line, frequency, schedule, priority, estimate and an optional checklist. Leads can also add their own rules on projects they lead.

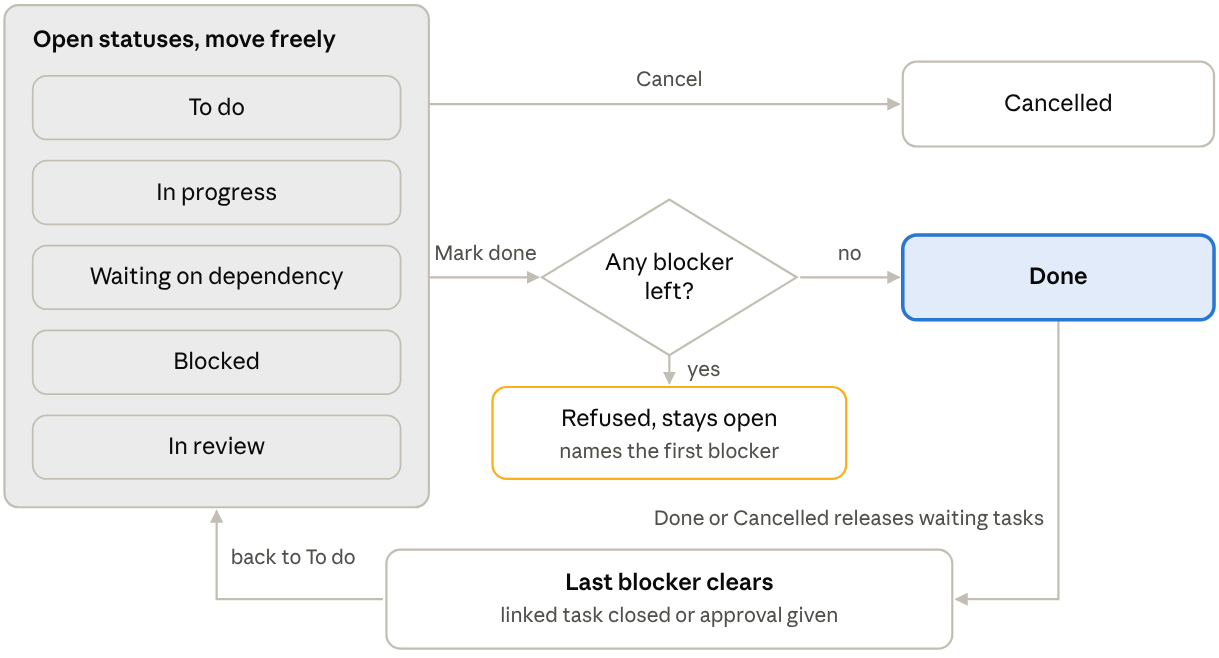
Move to live (clients)

When the admin moves a client to live, the app creates the project if needed, adds template rules for each service the client bought, and runs generate for that project immediately so the team sees tasks the same day.

Task lifecycle

A task moves freely between five open statuses and closes as Done or Cancelled, but it can only reach Done when nothing blocks it. Blockers are open linked tasks plus pending approvals, and clearing the last blocker moves a Waiting task back to To do automatically.

A task reaches Done only when no blocker remains



task status flow · 5 open statuses, 1 check, 2 closed

Mark done passes only when no linked task is open and no approval is pending; closing a task moves the tasks waiting on it back to To do.

Status	Code	Open or closed	Set by
To do	TODO	Open	Default for new tasks; automatic when the last blocker clears
In progress	IN_PROGRESS	Open	Person, or automatic on the first time log
Waiting on dependency	WAITING	Open	Automatic when a dependency or approval is added, or by hand
Blocked	BLOCKED	Open	Person, for blockers outside TaskDesk
In review	IN_REVIEW	Open	Person
Done	DONE	Closed	Person, refused while blockerCount is above 0; sets completedAt
Cancelled	CANCELLED	Closed	Person; sets cancelledAt; replaces deleting for non admins

Rules in setStatus(task, status)

1. If the new status is DONE and blockerCount(task) is above 0, refuse with the first blocker's name.
2. Set completedAt when DONE, cancelledAt when CANCELLED, clear both otherwise (reopening is allowed).
3. Write an activity "changed status" with "From to To".
4. When the task closes, run releaseDependents: every task that depends on it, is WAITING and now has no blockers, moves to TODO; its assignee gets an UPDATE notification "Ready to start".

Dependencies

- blockerCount(task) = linked tasks that are not closed + approvals still PENDING.
- Link candidates: open tasks in the same project (or the same assignee when there is no project), excluding the task itself and any task that already depends on it, so two way loops are impossible. For longer loops, check the chain on save and refuse a cycle.
- Removing the last blocker of a WAITING task moves it to TODO.

Approvals (pending approval from a person)

1. Anyone who can edit the task picks a person and writes what needs approval. A TaskApproval row is created as PENDING and the task becomes WAITING.
2. The approver gets an APPROVAL notification and sees Approve and Reject on their dashboard and on the task.
3. Only the approver or the admin can decide. Approve or Reject writes an activity and notifies the assignee.
4. When the last approval is approved and no task blockers remain, the task returns to TODO with the activity "dependency cleared". A rejected approval stays visible and keeps nothing blocked; the assignee decides what to do next.

Subtasks, collaborators, time

- Subtasks have their own assignee and done flag. The subtask assignee can tick it even without edit rights on the parent. Assigning a subtask notifies that person.
- Collaborators can edit the task and get an ASSIGNED notification when added.
- Time logs add minutes to the task total. Logging time on a TODO task moves it to IN_PROGRESS.
- Comments support @mentions (see Notifications).

Task page layout (compact, approved in Version 9)

- Header: project, cadence chip, rule description, title, then status pill, a Move to dropdown, a Mark done button and an edit icon that opens the Edit task popup.
- Main column: one card for details and subtasks, one card for discussion, history collapsed by default.
- Side card (300 px max): Waiting on (link a task or ask for approval), People, Log time on one line, created details.
- Deleting is only inside the Edit popup and only for the admin, with a two click confirm.

Calendar, holidays and leave

A working day is any day that is not a weekly off and not a public or company holiday. The default weekly off is every Sunday plus the 2nd and 4th Saturday; only the admin approves leave.

Working day logic (src/lib/calendar.ts)

- isWeekend(day): Sunday when sundayOff is on; Saturday depends on satRule: ALL_WORK, SECOND_FOURTH (the week number is the day of month divided by 7, rounded up, and it is off when that is 2 or 4) or ALL_OFF.

- holidayOn(day): a Holiday of type PUBLIC or COMPANY on that date. RESTRICTED holidays are not company wide days off.
- isWorking(day) = not isWeekend and not holidayOn.
- workingDays(from, to) counts working days in a range; leave days use it, so a leave across a Sunday counts only working days. A half day counts as 0.5.

Holidays

- The admin keeps the list (the Search Engineers calendar from December 2025 to December 2026 is the seed). Each holiday has a date, a name and a type.
- Adding, removing or changing a holiday, or changing the weekly off settings, calls rescheduleAll so future untouched tasks move to the right dates.
- Restricted holidays: each person may pick up to rhLimit (2) per calendar year. Picking one creates a Leave of type RH that still needs admin approval.

Leave flow

1. A person requests leave: type, from, to, half day (single day only), reason.
2. Checks on save: no overlap with their own pending or approved leave; the yearly allowance for that type is not exceeded (pending plus approved plus this request). Allowances are settings: Casual 12, Sick 12, Earned 15; Comp off and Unpaid have no cap.
3. Status PENDING; every admin gets a LEAVE notification. An admin's own leave is recorded as APPROVED directly.
4. The admin approves or rejects. The person gets a LEAVE notification and an activity is written.
5. The person can cancel a pending or future approved request.
6. Approved leave shows on the calendar, on the Team page, on People cards and in the assignee picker as "On leave".

Calendar page

A month grid showing weekly offs, holidays (public and restricted), approved team leave and task due counts per day. The admin edits holidays and weekly off rules from here.

Attendance, activity log and profiles

Attendance comes from sign in and sign out sessions, and every meaningful action writes one Activity row; a person's profile shows both as one timeline that anyone in the team can read.

Sessions

1. Successful sign in: if the person has an open session from today, reuse it; if the open session is from an earlier day, close it at 18:30 that day; then open a new Session with signedInAt = now.
2. Sign out: close the open session. A session from an earlier day closes at 18:30 of that day, never later.
3. The daily job closes any session left open from an earlier day.
4. dayAttendance(person, day) = first sign in, last sign out, total minutes across that day's sessions, number of sessions, still open or not.
5. Attendance page: leads and admin see a month table per person, members see their own, with first in, last out and hours; CSV export.

Activity log

- Write it with act(action, detail, taskId, projectId) from inside the same transaction as the change.
- Actions used today: created task, changed status, completed task, edited task, added or removed subtask, assigned subtask, added or removed dependency, dependency cleared, requested approval, approved, rejected, added collaborator, logged time, commented, deleted task, requested leave, cancelled leave, approved leave, rejected leave, read policy, updated policy, added holiday, removed holiday, changed weekly off, updated leave allowances, created project, changed project status, updated project members, added, edited, paused, resumed or deleted recurring task, added template recurring tasks, added team member, shared announcement, added client, moved client, moved client to live, renewed, renewal reminder created.
- Keep all rows in production (no cap). Add indexes on (userId, at) and (taskId, at).

Profile page (people/[id])

- Profile pages (people/[id]) show details, four figures (open tasks with overdue, done this month, logged this week, today's status), Working on, Completed in the last 30 days, projects and direct reports. They no longer contain the log; an Activity log button appears only when the viewer may see that person's log.
- The log lives on its own page, /activity, in the main sidebar group. canSeeLog(viewer, person): admin sees everyone; a lead sees themselves and every Member; a member sees only themselves. The person picker lists only allowed people, with Everyone (admin) or Me and all members (lead) as a combined view; members get no picker.

- Log entries = Activity rows for the selected people (minus sign in, sign out and commented) + one entry per Comment + one sign in and one sign out entry per Session, merged newest first and grouped by day. In the combined view each line starts with the person's name, which filters to that person.
- Figures for the selection and period: entries, tasks completed, time logged, days signed in. Side card: per person entries, done and time (combined view) or a summary for one person.
- Filters: category tabs with counts (All, Tasks, Comments, Time, Attendance and leave, Other), period (today, 7, 30, 90 days, all time), text search; 80 entries per page with Show more. Filters reset when a different person signs in.
- Category rule: commented = Comments; logged time = Time; sign in, sign out and anything with leave = Attendance and leave; anything about tasks, subtasks, status, dependencies, collaborators or approvals = Tasks; the rest = Other.
- Client actions and leave allowance changes are shown only to the admin.
- The Team page Recent activity feed follows the same canSeeLog rule.

Notifications, mentions and announcements

Every notification is one row for one person, created by notify(userId, kind, title, body, link) inside the action that caused it; the bell in the sticky top bar shows that person's unread count. Never notify the person who did the action.

Kind	Sent when	Sent to	Opens
ASSIGNED	A task is created for or reassigned to someone; added as collaborator; subtask assigned; recurring tasks generated (grouped)	The assignee or collaborator	The task
MENTION	A comment contains @Name	Each mentioned person	The task
APPROVAL	Approval requested; approval decided	The approver; then the assignee	The task
LEAVE	Leave or restricted holiday requested; leave decided	All admins; then the requester	Leave page

Kind	Sent when	Sent to	Opens
RENEWAL	A domain, hosting or AMC renewal is 30 days away	The client owner	The reminder task
POLICY	A policy is updated	Everyone who must read it again	Policies
ANNOUNCE	The admin shares an announcement	Everyone active except the author	The announcement
UPDATE	A task becomes ready to start; a comment on your task; you were added to TaskDesk; a person now reports to you	The affected person	The related page

Mentions

1. The comment box accepts @ followed by a full or first name; a Mention dropdown inserts "@Full Name " for people who prefer picking.
2. On save, match names against active people (longest names first, case insensitive, whole words) and store the comment as typed.
3. Each mentioned person gets one MENTION notification. If the task assignee was not mentioned, they get an UPDATE instead.
4. Rendering highlights each mention in red. Escape the comment text before highlighting.

Bell, page and read state

- Bell: red count pill (99+ cap); clicking opens the latest 7 with Mark all as read and See all notifications.
- Notifications page: tabs All, Unread, Mentions, Tasks, Approvals, Announcements with unread counts; each row has mark read or unread. Opening a notification marks it read and goes to its link.
- Poll the unread count every 60 seconds, or push it with server sent events later.

Announcements

- Admin writes a title and body from the Notifications page (New announcement). Saving creates the Announcement and one ANNOUNCE notification per active person.

- The newest unread announcement from the last 30 days shows as a red banner under the top bar for that person. Bodies over 150 characters are cut at a word with Read more, which opens a popup.
- Mark as read (the cross, or the popup button) writes AnnouncementRead and marks the notification read, so the banner disappears for that person only.

Clients, services and renewals

Clients are admin only. A client moves through eight stages, holds one ClientService row per service it buys (each with its own price, type and dates), and only the live stages Active and Onboarding count toward revenue.

Stages

Lead, Proposal sent, Negotiation, Onboarding, Active, On hold, Completed, Churned. Changing the stage writes a ClientNote ("Stage changed from X to Y") and an activity. Reaching Onboarding or Active for the first time sets onboardedOn.

Services and pricing

- Lines: SEO, Google Ads, Content and social, Web development, Website AMC, Reporting and analytics, Domain, Hosting, Other services.
- Type RECURRING has a cycle (MONTHLY, QUARTERLY, YEARLY); type ONE_TIME has a single price. One client can mix both, for example a one time website plus monthly SEO.
- Each service has startDate and completedDate. A service is running when it has no completedDate or the completedDate is today or later. "Mark completed" sets completedDate to today.
- Domain, Hosting and AMC rows also hold item (for example the domain name), provider and renewDate.

Dashboard formulas

Figure	Formula
Active clients	Clients in Active or Onboarding
Monthly recurring	For live clients, running RECURRING services started by month end, converted to a month: monthly as is, quarterly divided by 3, yearly divided by 12

Figure	Formula
One time this month	ONE_TIME services whose startDate is in this month, for clients past Negotiation
Monthly total (projected)	Monthly recurring + one time this month, shown with both parts
Upcoming renewals	Domain, hosting and AMC renewals in the next 30 days + contract renewals of live clients in the next 45 days
Ongoing clients	Live clients out of all clients ever added
Onboarded this month	Clients whose onboardedOn is in this month
Received this week and this month	Sum of Payment.amount from Monday (or the 1st) to today
Still to collect	Projected minus received this month, never below zero
Services we provide	Per line: number of clients with a running recurring service, its monthly value, number and value of one time projects

Ad spend managed is not tracked.

Renewal reminders (renewalReminders, daily job)

1. For every client that is not Churned, find running Domain, Hosting and AMC services whose renewDate is between today and 30 days ahead.
2. Build a key from the service id and its renewDate. If a task with that renewKey exists, skip.
3. Create a task for the client owner on the client's active project: "Renew <domain, hosting or website AMC> <item> for <client>", due 30 days before renewal (or today if that has passed), priority URGENT within 7 days, otherwise HIGH, with subtasks Confirm with the client, Renew, Update the renewal date.
4. Send a RENEWAL notification and write the activity "renewal reminder created".
5. Clicking "Renewed for 1 year" on the client moves renewDate forward one year and logs a note; next year a new key creates a new reminder.

Move to live

The admin picks project name, service line, lead, members, start date and whether to add template rules. The app creates the project (copying website, industry, POC and comms), adds template rules for each recurring service and one time rules for each one time service, sets the stage to Active, fills onboardedOn and missing service start dates, writes a client note, then runs generate for that project.

Contacts and communication

Each client has a point of contact (name, position, phone, email), any number of other contacts, and communication preferences chosen from Phone call, WhatsApp group, Email and Direct visit. Projects show the same POC and preferences so the team knows how to reach the client.

Payments

The admin records amount, date and note. Payments feed the received figures; there is no invoicing in version 1.

Credentials, tools and security

Passwords for client and internal logins are encrypted at rest, sent to the browser only when a person allowed to see them clicks the eye or copy icon, and every reveal is logged. The preview stores them in plain browser storage; production must not.

Credential access

Level	Code	Lead or member sees
Hidden	NONE	Not listed at all
Login only	LOGIN	Name, link and username, no password
Full access	FULL	Everything, including the password

- Each credential has its own leadAccess and memberAccess, set inline by the admin on the list.
- Admin always has full access. Tabs: Client (grouped by client) and Internal.
- List query: admin gets all; others get rows where their role's access is not NONE. Never select passwordEnc in list queries.

- Reveal and copy call a server action that checks credLevel is FULL, decrypts, returns the single password, and writes CredentialView. The admin sees an access log under the list.
- Edits open in a popup modal (name, client, link, username, password, notes, access). Saving a new password sets changedOn.

Tools list

- Fields: Name, Category (SEO, Analytics, Advertising, Content and design, Development and hosting, Communication, Productivity, Others), Login link, Login (username, email or mobile), Password, 2FA details, Free or Paid, Notes.
- Access type: Own login, Shared login (ask admin), Free with no login. Password visibility: admin only, leads, or everyone.
- Everyone sees the list and links; the password follows the same reveal and log rule as credentials.

Security checklist

1. Encrypt passwords with AES 256 GCM using a key from an environment variable (CREDENTIALS_KEY, 32 bytes). Store iv, tag and ciphertext together.
2. Hash user passwords with bcrypt (cost 12). New users get a temporary password and mustChangePassword = true.
3. Session cookie: httpOnly, sameSite lax, secure when served over HTTPS (COOKIE_SECURE), signed with AUTH_SECRET, 7 day expiry.
4. Rate limit sign in: 5 failed tries per email per 15 minutes.
5. Every server action checks permissions server side, including hidden fields such as role and reportsTo.
6. Escape all user text before rendering; comments render mentions after escaping.
7. Back up the database daily and keep 14 days. The CREDENTIALS_KEY is backed up separately from the database.

Dashboard and reports

My dashboard is the home page for every role: simple widgets that summarise each section and a Plan my day widget that fills an 8 hour day. Reports measure tasks due in a period against tasks done, by day, week, month or quarter.

Widgets (in order)

Widget	Shows	Who
Plan my day (full width)	Morning, Afternoon and Evening slots, minutes left of 480, Plan for me, add, move or remove tasks	Everyone
Today	Overdue first (top 3), tasks due today, subtasks assigned to me on others' tasks	Everyone
This week	Open tasks due in the next 7 days (top 7)	Everyone
Notifications	Latest unread	Everyone
Waiting for your approval	Task approvals with Approve and Reject; pending leave count for the admin	Only when something waits
Attendance and leave	Signed in time, time today, leave left per type	Everyone
Holidays and team leave	Holidays in the next 30 days, team leave in the next 14	Everyone
My projects	Open and overdue counts per project	Everyone
This month	Percent of my due tasks done, done count, overdue, time logged	Everyone
Policies to read	Unread policies with a Read button	Only when unread
Clients	Active clients, monthly recurring, pipeline, renewals	Admin
Quick links	Tool links	Everyone
Quick add	Title, project, due date, priority	Everyone

Plan for me: take my open tasks (as assignee or collaborator) due today or earlier that are not planned yet, sorted by priority (Urgent first) then due date. Add each using its estimate or 30 minutes while the running total stays within 480 minutes, skipping any that would not fit. The slot follows the running total: up to 210 minutes is Morning, up to 420 is Afternoon, the rest is Evening. The plan is saved per person per date and starts empty each day; Add to plan offers tasks due within the next 3 days.

Reports

- Period: Daily, Weekly, Monthly, Quarterly with an anchor date; filters by project and person. A period that includes today counts only up to today.
- Figures: Tasks due (and done, cancelled), Completion (done of due, green at 90 percent or more, red under 70), On time (completed on or before the due date), Still overdue, Time logged, One time tasks (added outside routines).
- Breakdowns: per person, per project and per cadence; a bar chart of tasks completed each day (each week for a quarter); a list of tasks due in the period and still open.
- CSV export via api/reports/csv with the same filters. Members see their own numbers; leads and admin see everyone.

Pages, routes and UI conventions

The app has 20 signed in pages under one shell: a collapsible sidebar on the left and a sticky top bar with the page title, date, New task, the bell and the avatar. Build each page from the matching preview screen.

Route	Page	Who
/	My dashboard	Everyone
/tasks, /tasks/new, /tasks/[id]	Task list with filters, new task, task detail	Everyone
/team	Who is in, overdue per person, recent activity	Everyone
/projects, /projects/new, /projects/[id]	Project list with industry, line and status filters; project with rules, members, POC	Everyone; create for leads and admin
/leave	Requests, balances, restricted holidays, admin approval queue	Everyone
/reports	Period reports	Everyone, own numbers for members
/calendar	Month view of offs, holidays, leave	Everyone
/attendance	Monthly attendance	Everyone; members see only their own rows

Route	Page	Who
/people, /people/[id]	Directory and profiles (no log on profiles)	Everyone
/policies	HR and Leave policies with acknowledgement	Everyone; edit for admin
/tools	Tool list	Everyone
/credentials	Client and Internal tabs	Admin, plus anything shared
/clients, /clients/[id]	Client dashboard and client page	Admin
/notifications	All notifications and announcements	Everyone
/login	Sign in	Public

Sidebar

- Always visible: My dashboard, Tasks, Projects, Team, Leave, Reports. A View more button reveals two groups under red rulers with white text: Office (Calendar, Attendance, People, Policies, Tools, Credentials) and Admin (Clients).
- The panel icon collapses the sidebar to a 64 px icon rail; the choice is remembered per browser. Under 820 px the sidebar becomes a drawer opened from the top bar menu icon.
- Red badges: pending leave for the admin, unread policies for everyone.

Look and feel

- White background, black primary buttons that turn brand red on hover, brand red for links, active nav and accents. Outfit for headings, Figtree for text.
- Every edit form for an existing record opens in a popup modal (credentials, tools, rules, people, tasks).
- Destructive actions use a two click confirm on the same button, never a browser confirm dialog.
- No page may scroll sideways: wide tables scroll inside their own box, grid children get min width 0, the main area clips horizontal overflow. Check at 390 px, 1000 px, 1280 px and 1600 px.

- Dates show as "Sun, 11 Oct" or "11 Oct 2026"; money as Indian rupees with Indian digit grouping.

Tasks page: Board (default) and List (approved in Version 12)

- A Board and List switch sits in the page header. Board is the default; the choice is remembered per person.
- Board columns: To do, In progress, Waiting on dependency, Blocked, In review, Done. With the Open filter, Done shows tasks completed in the last 14 days; Any status adds a Cancelled column; picking one status shows only that column.
- Cards show title, project, cadence, due date, Waiting chip, priority, subtask and comment counts and the assignee avatar. The left border colour follows priority (Urgent red, High amber).
- Drag a card to another column to change status, using the same setStatus rules (Done is refused while blockers remain). Only people who can edit the task can drag it. On touch screens each card shows a Move to dropdown instead.
- Columns show 6 cards with Show all N. Six columns fit side by side on a laptop; narrower screens scroll the board sideways inside its own box, never the page.
- The filter bar (search, person, project, status, cadence, overdue only, reset) is one compact row shared by both views.

Read more rule

- Any text block over 6 lines (task details, comments, project scope, log details) is cut at 6 lines with a Read more link that toggles to Show less.
- Any list over 6 items (open tasks, recurring rules, contributors, team, activity feeds, profile lists) shows the first 6 with Show all N.
- In the preview this is one helper, applyClamps(), run after every render on elements marked data clamp text or list. Build it as one shared React component used everywhere.

Project page (approved in Version 11)

- Header like a profile: initials badge, name, status and service line chips, client, industry, lead, team size, website; Report and New task buttons.
- Period switch: This week, This month (default), This quarter, All time.
- Six figures: tasks due in the period (and so far), completion (done of due so far, green at 90 percent, red under 70), on time, open now with overdue, time logged in the period, planned hours per month from active rules.

- Tasks by cadence: Daily, Weekly, Monthly, Quarterly, One time, each with its rule count, due, done, open now with late count, completion bar and time logged.
- Done each month: the last 6 months, done against due, as paired bars.
- Who contributed: everyone who leads, is a member, was assigned, collaborated or logged time; role, tasks done in the period, share of done as a bar, open now with late count, time logged, comments. Sorted by done, then time.
- Then recurring rules, open tasks, client contact, team, recent activity, scope and notes, and project settings for the lead and admin.

Background jobs, API and deployment

One daily job keeps the app current: run it at 00:05 Asia/Kolkata, and it is safe to run again because every step skips work already done.

Daily job (POST api/cron/generate with header Authorization: Bearer CRON_SECRET, or npm run generate:tasks)

1. Close sessions left open from earlier days at 18:30 of their day.
2. generate() for all active rules with the 7 day lookahead.
3. renewalReminders() for domains, hosting and AMC due in 30 days.
4. Return counts as JSON (tasks created, reminders created, sessions closed) and write one system activity.

Also call generate for one rule or project right after a rule or project is created or edited, and rescheduleAll after holiday or weekly off changes, so nobody waits for the night job.

Endpoints

Endpoint	Method	Purpose	Auth
api/cron/generate	POST	Daily job	CRON_SECRET
api/reports/csv	GET	Report export with period, anchor, project and person	Signed in, same rules as the Reports page
api/notifications/count	GET	Unread count for the bell	Signed in
api/attendance/csv	GET	Month attendance export	Leads and admin, members for their own

Everything else is a server action next to its page.

Environment variables

Name	Example	Purpose
DATABASE_URL	file:./taskdesk.db or a postgresql URL	Database
AUTH_SECRET	64 random hex characters	Signs session cookies
CRON_SECRET	random string	Protects the daily job
CREDENTIALS_KEY	32 random bytes, base64	Encrypts tool and credential passwords (new)
APP_TIMEZONE	Asia/Kolkata	Defines today
GENERATION_LOOKAHEAD_DAYS	7	Days of tasks created ahead
ADMIN_NAME, ADMIN_EMAIL, ADMIN_PASSWORD	Admin, admin@searchengineers.in, a strong password	First admin created by the seed
SEED_DEMO	false	Load demo team and projects
COOKIE_SECURE	true behind HTTPS	Secure cookie flag

Deploy on one server

1. Install Node.js 20 or newer. Copy the project, then npm ci.
2. Create the env file from the example above, then npx prisma db push (or prisma migrate deploy once migrations exist) and npm run seed.
3. npm run build, then run npm start under a process manager (pm2 or systemd) behind Nginx with HTTPS.
4. Add a cron line at 00:05 Asia/Kolkata calling the daily job with the secret.
5. Seed the holiday list and the HR and Leave policies from the preview, then sign in as admin and add the team.

SQLite is fine for this team size on one server. Move to PostgreSQL when more than one app server is needed or the database passes a few GB.

Build order and testing

Build in six phases, each usable on its own, so the team can start using TaskDesk after phase 2 while the office and client modules follow. Each phase ends when its checks below pass at desktop and phone width.

Status on 11 October 2026: all six phases are built in the taskdesk source (Next.js 15, Prisma 6, 30 tables, about 7,300 lines). The production build passes and 39 end to end checks pass across Admin, Lead and Member at desktop and phone width. The daily job runs with `npm run daily` or the cron endpoint. The code layout and patterns are in docs/CONVENTIONS.md inside the source.

1. **Foundation**: schema for all models, permissions.ts, calendar.ts, act() and notify(), the app shell (sidebar with View more, sticky top bar, bell), people management with Add lead or member and Add member, profile pages.
2. **Tasks and recurrence**: projects, rules and templates, the generator and daily job, task list and the compact task page with subtasks, dependencies, approvals, comments with mentions, time logs. Team page.
3. **Dashboard and notifications**: My dashboard widgets and Plan my day, notifications page, announcements banner.
4. **Office**: calendar, holidays, weekly off settings, leave and restricted holidays, attendance, policies with acknowledgement, tools.
5. **Admin**: clients with services, payments, renewals and move to live; credentials with encryption, access levels and the view log.
6. **Reports and launch**: reports and CSV, attendance export, data import of the real holiday list, policies and team, backups, go live.

Porting from the preview

- The preview is one HTML file. Its logic functions map almost one to one to src/lib: isWeekend, holidayOn, isWorking, workingDays, leaveDays, matches, occurrences, scheduleDate, generate, resetRuleFuture, rescheduleAll, setStatus, releaseDependents, blockerCount, decideApproval, renewalReminders, recurringMonthly, oneTimeThisMonth, personLog, dayAttendance.
- Replace the in memory state with Prisma queries, and each data act or data form handler with a server action of the same name.

- Keep the labels, status names and messages exactly as the preview shows them; the team has already reviewed that wording.

Acceptance checks

- [ ] A weekly rule on a project creates exactly one task per week up to 7 days ahead, and running the job twice creates nothing new
- [ ] A monthly task falling on a holiday moves to the next working day and shows "Moved from"; a daily task on a holiday is skipped
- [ ] Changing the Saturday rule or adding a holiday moves untouched future tasks and leaves started ones alone
- [ ] A task with an open dependency or pending approval cannot be marked done; clearing the last blocker moves it from Waiting to To do and notifies the assignee
- [ ] An @mention notifies only the mentioned people, and the bell count updates
- [ ] A lead can add only members, can edit only members who report to them, and the server refuses a forged role
- [ ] Only the admin can approve leave, delete tasks, see clients and set credential access
- [ ] A member with Login only access never receives the password from any request; every reveal is logged
- [ ] Renewal reminders appear 30 days before a domain, hosting or AMC date and are never duplicated
- [ ] Profile logs show tasks, comments, time, sign in and sign out; client entries are hidden from non admins
- [ ] No page scrolls sideways at 390, 1000, 1280 and 1600 px
- [ ] Report numbers for a closed month match a manual count from the database